Supplementary Information for: SARS-CoV-2 Delta variant re-emerges in US farmed mink and free-ranging white-tailed deer in 2022–2023

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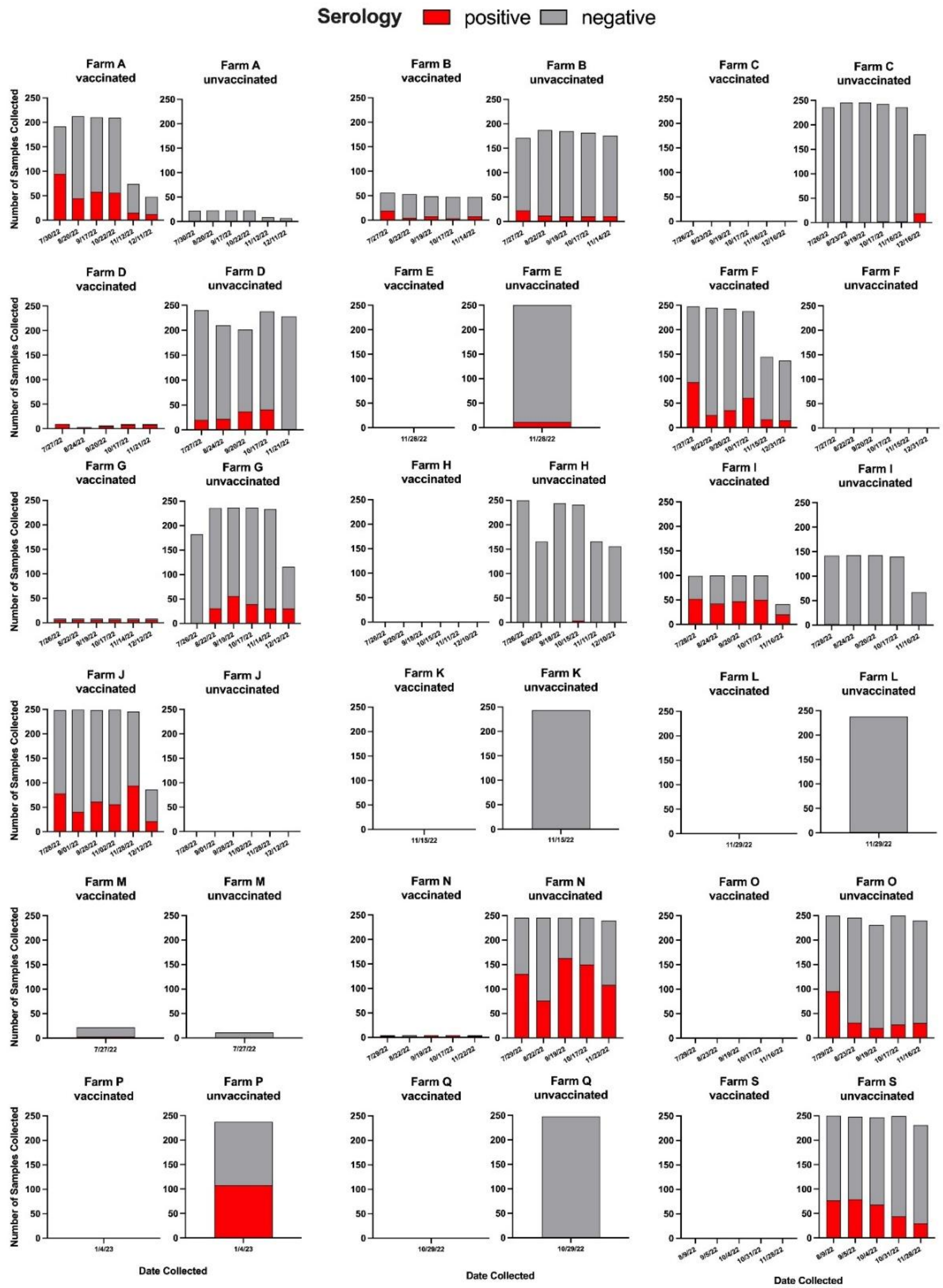

Figure S1. SARS-CoV-2 antibodies across U.S. mink farms by collection date. The number of seropositive (red) mink and seronegative (gray) mink by vaccination status is shown at each collection date, highlighting farms with potential seroconverted animals. To identify farms with natural SARS-CoV-2 introductions for virological follow-up, we used serology results from an ELISA targeting the B.1.1.529 (Omicron) spike protein, which showed an overall seropositivity of 18.1% (2,884/15,903). We focused on farms with primarily unvaccinated mink to account for preexisting immunity from widespread SARS-CoV-2 vaccination of mink in 2021. Farms N and O were prioritized for testing due to high seroprevalence in the first month of the study, 53.6% (134/250) and 38.0% (95/250), respectively. Farm S was included due to its epidemiological link to Farms N and O and a high seroprevalence 42.0% (105/250) during the study. Farm P, another unvaccinated farm where 45.6% (108/237) of mink were seropositive by the study's end. A follow-up study was also conducted in Farm P on April 2024 to assess whether there was continued circulation of SARS-CoV-2 on Farm P. A total of 9/50 sampled mink tested positive for SARS-CoV-2 antibodies at that time. Source data are provided in a Source Data file.

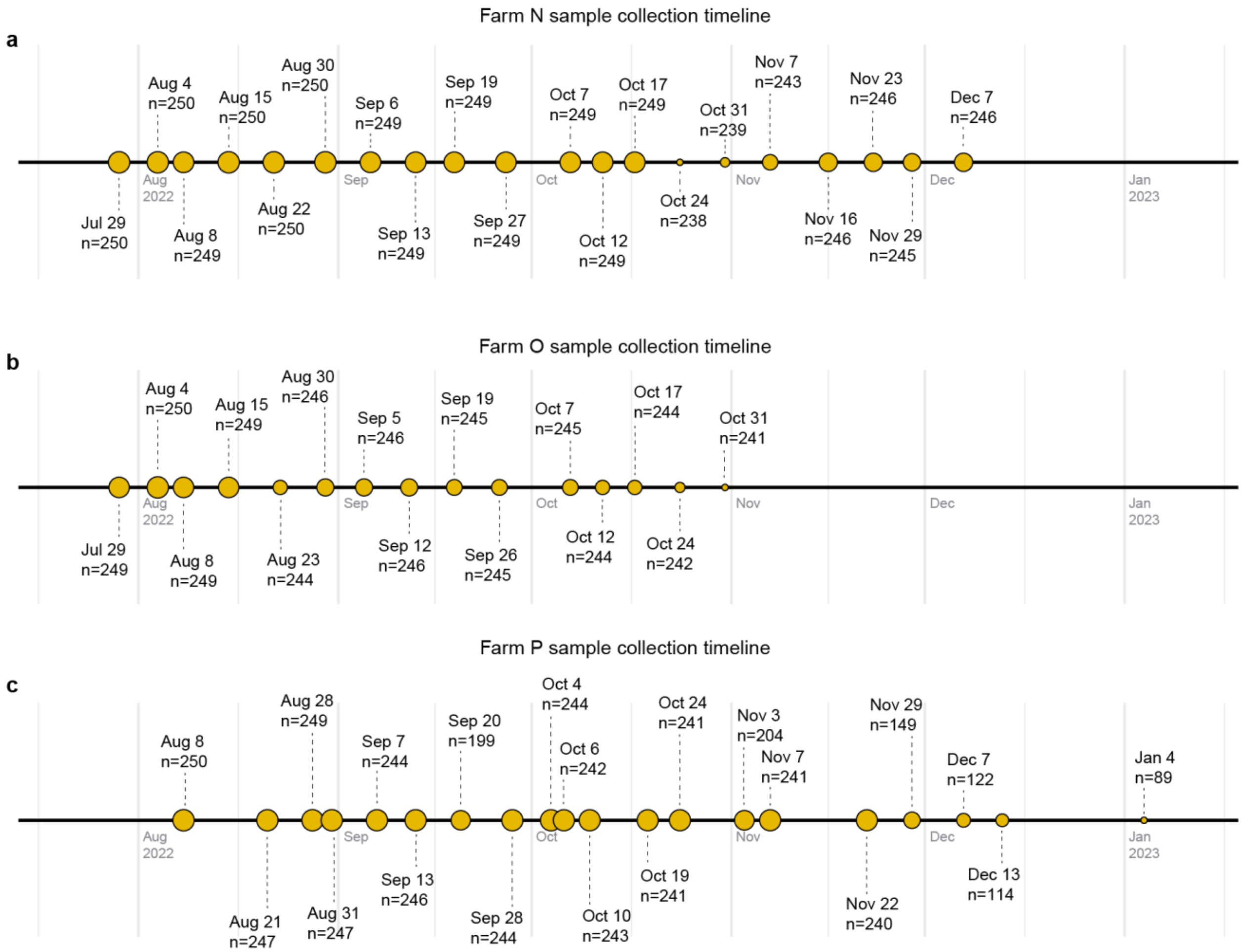

Figure S2: Timeline of sample collection. The number of samples collected are provided for each collection date for (a) Farm N from State 1 (n=4,945) (b) farm O from State 1 (n=3,685) and (c) Farm P from State 2 (n=4,296). Circles indicate collection dates and are sized according to the number of samples collected on that date for each farm.

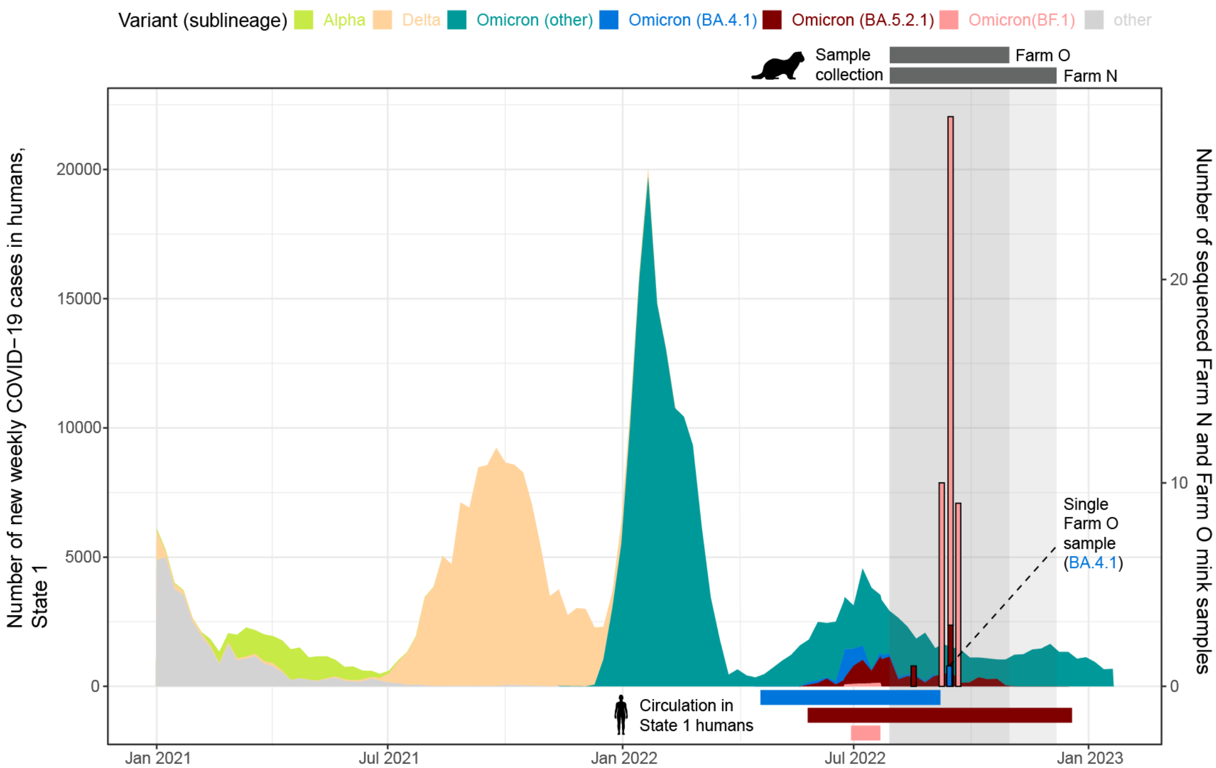

Figure S3: Contemporary circulation of Omicron sublineages sampled in humans and State 1 farmed mink. Epidemiological curves of the number of new weekly COVID-19 cases in State 1 humans from January 1, 2021 to January 31, 2023 and number of sequenced SARS-CoV-2 farm N (n=48) and farm O (n=1) mink samples. The curves are colored by the estimated proportion of State 1 human SARS-CoV-2 sequences belonging to one of three WHO-designated variants of concern (or 'other'). The gray, shaded boxes correspond to the sample collection periods. Horizontal bars above the plot show the collection date ranges for each farm. Horizontal bars below the plot show the circulation times for the BA.4.1, BA.5.2.1 and BF.1 Omicron sublineages in State 1 humans. Source data are provided in a Source Data file.

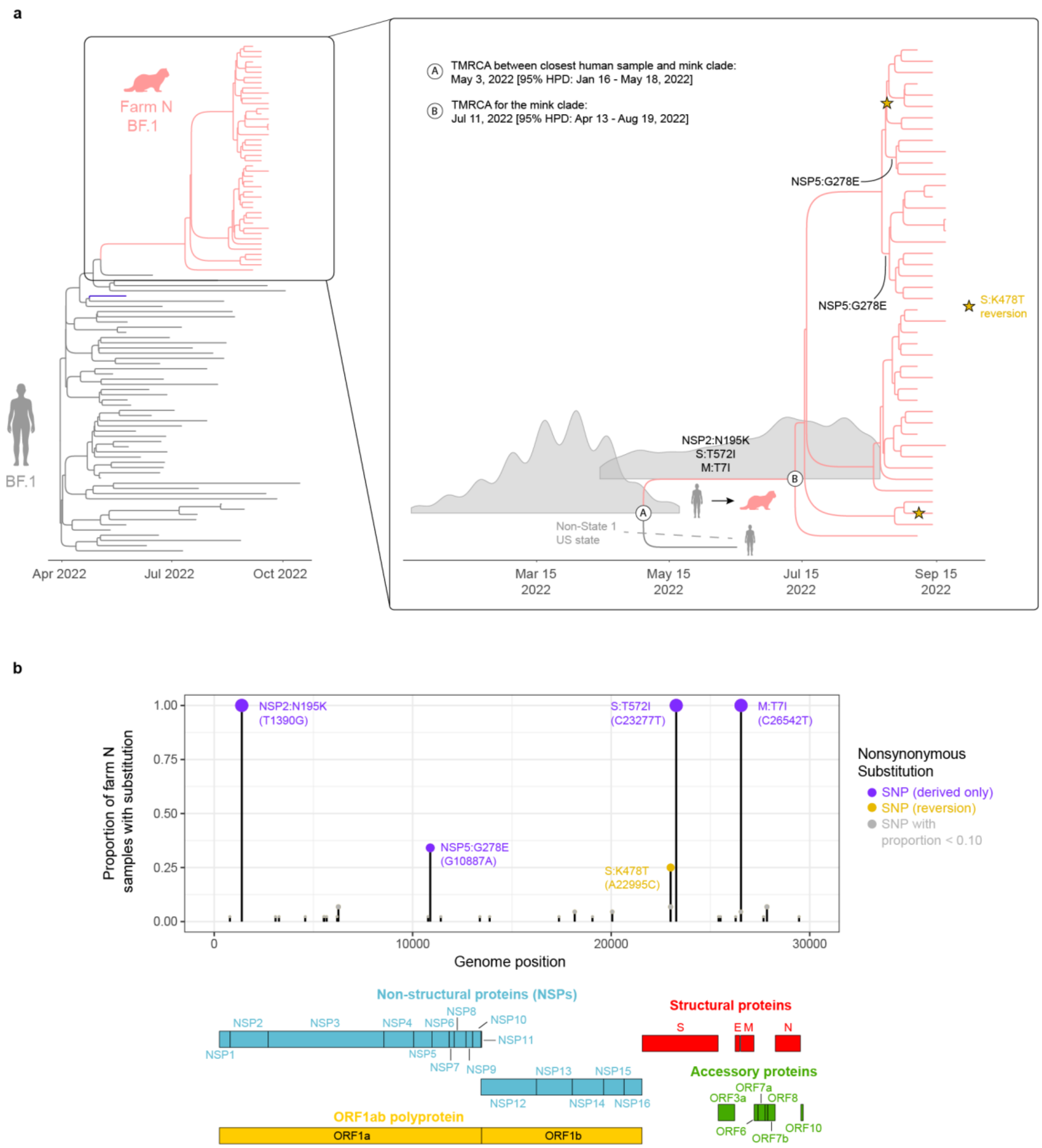

Figure S4. Evolution of BF.1 viruses in mink. (a) Time-scaled MCC tree of BF.1 viruses sequenced from Farm N mink (n=44) and humans (n=54). Branches are shaded by host: gray=human, pink=mink. A subsection of the MCC tree with the mink clade is shown in greater detail on the right. The 95% HPD density distribution plots for the TMRCA are shown for two key nodes: the common ancestor of the mink clade (node A) and the common ancestor of the mink clade and most closely related human virus (node B). Select clade-defining mutations are indicated on the branch where they first emerged. Spike reversion K478T is denoted by a star. (b) Proportion of Farm N samples with acquired substitutions since the time to the most recent human-sampled ancestor. Genomic coordinates of annotated SARS-CoV-2 open reading frames correspond to the Wuhan-Hu-1 reference genome sequence (GenBank RefSeq accession NC_045512).

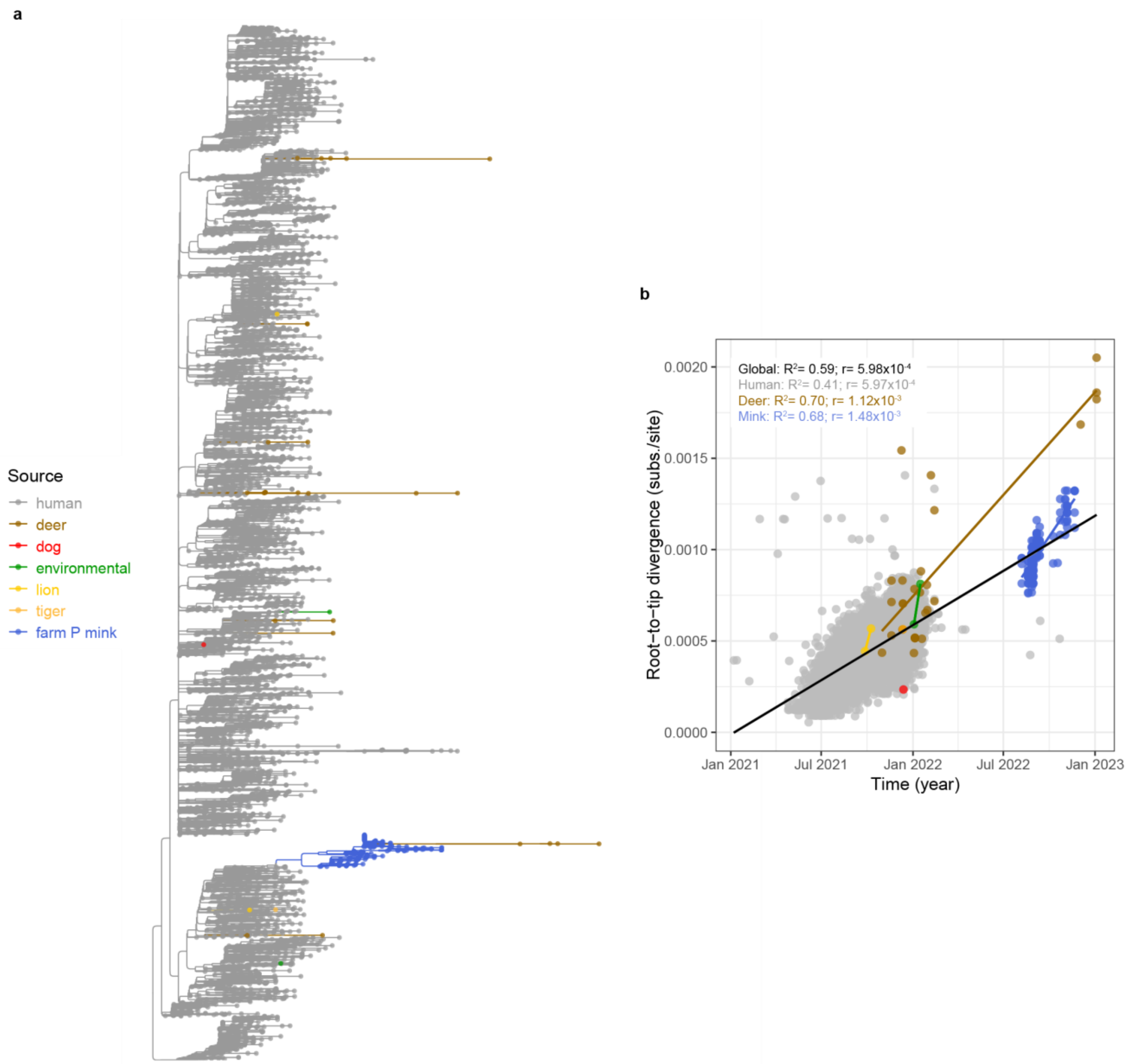

Figure S5: Global AY.39 phylogeny including Farm P mink sequences. (a) Midpoint-rooted maximum likelihood phylogenetic tree of complete genome AY.39 sequences collected from Farm P mink (n=244) and AY.39 sequences from GISAID that includes all available non-human-derived sequences collected from the environment (n=2), deer (n=28), dog (n=1), lion (n=2) and tiger (n=1) samples plus a random subset of human sequences (n=7,778). The subsampling strategy resulted in one randomly-selected genome for each collection date per US state, plus one randomly-selected genome for each collection date per country outside the US. (b) Root-to-tip regression analysis for the AY.39 phylogeny shown in (a). The slope of the regression *r* corresponds to the evolutionary rate for three main groups (human, deer, mink) under the assumption of a strict molecular clock. Source data for panel b are provided in a Source Data file.

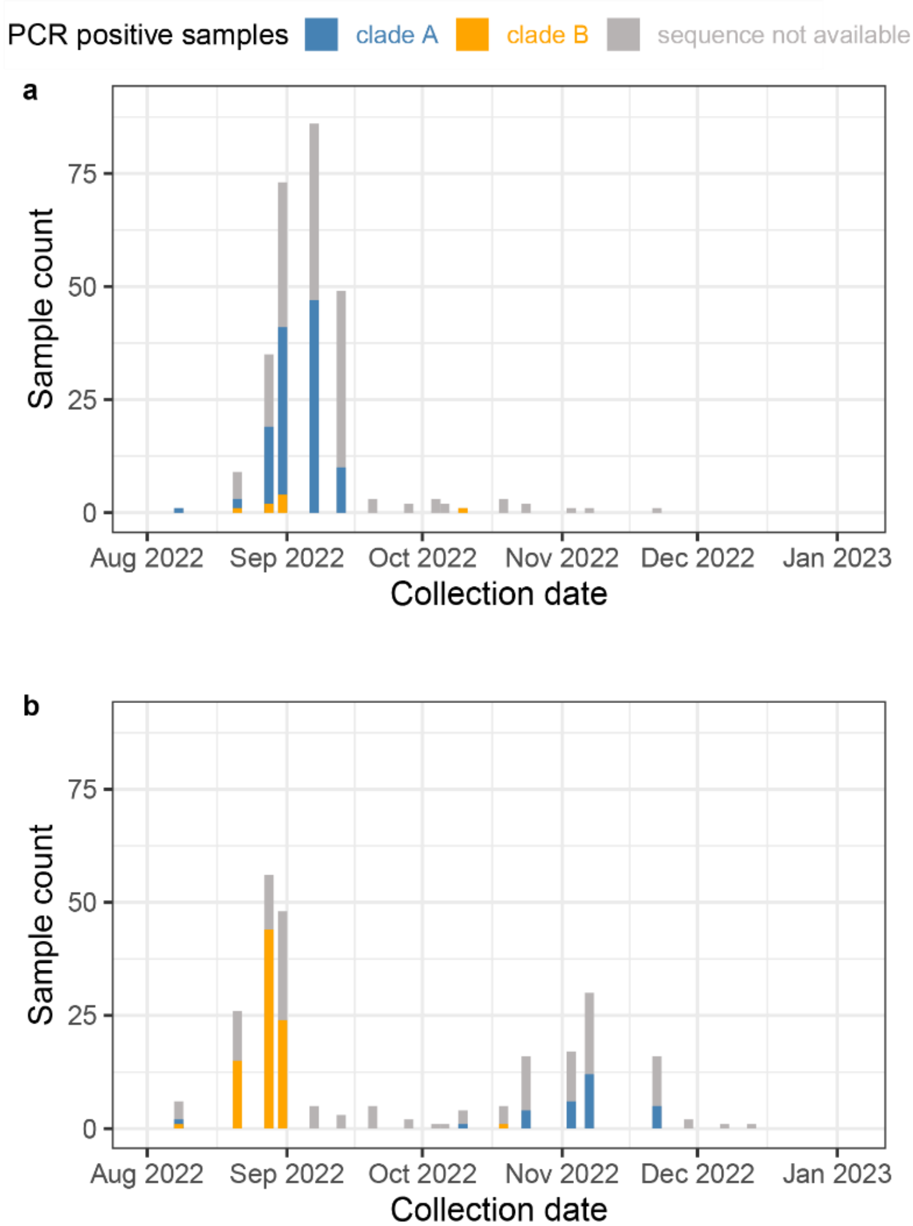

Figure S6: Number of rRT-PCR positive nasal swab samples and clade designation for sequenced isolates collected from farm P (a) barn P1 and (b) barn P2. Source data are provided in a Source Data file.

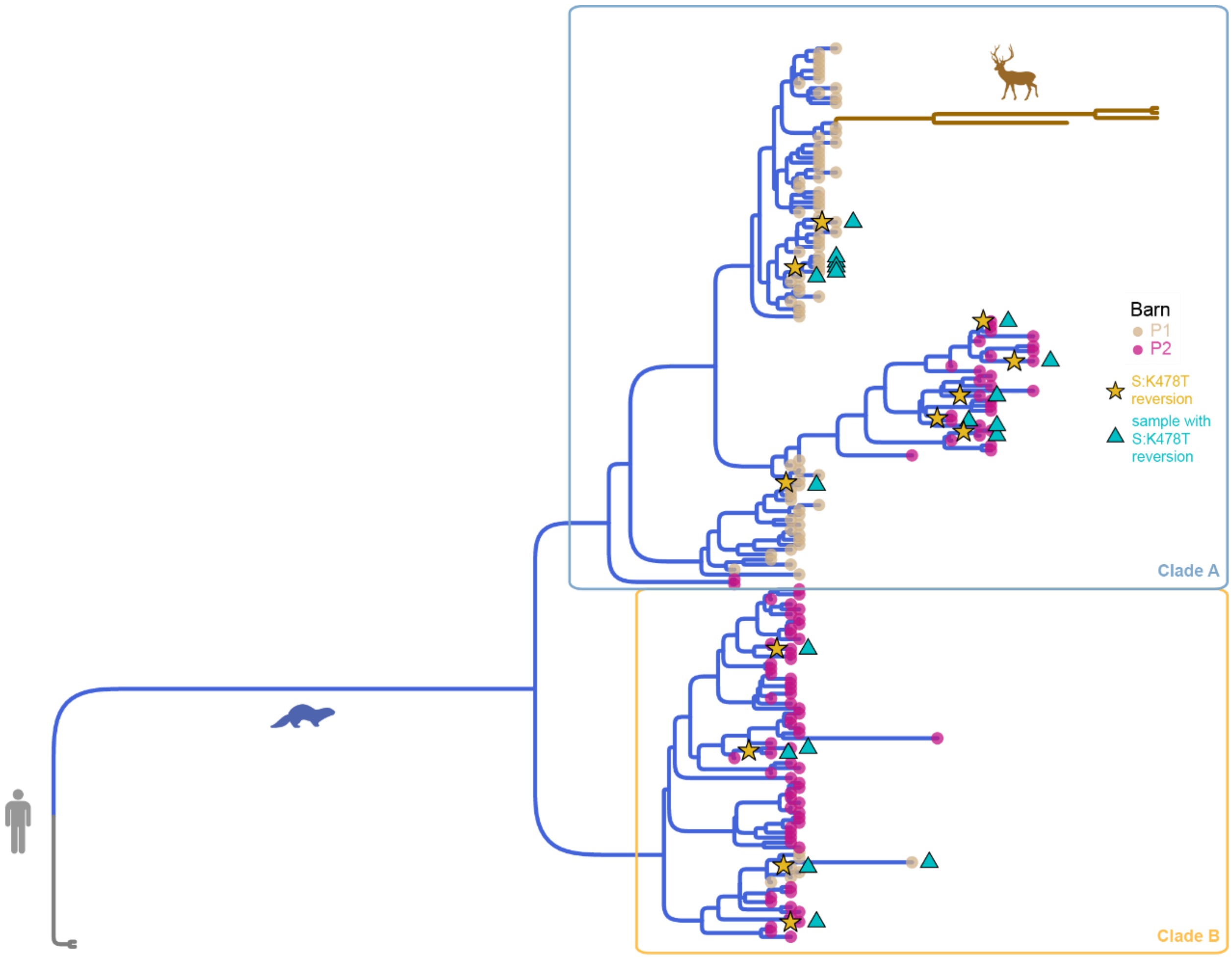

Figure S7. AY.39 phylogeny from Figure 5a depicting the independent emergence of spike reversion K478T. Branches are shaded by host: gray=human, blue=mink, brown=deer. Tips are colored by barn: yellow=P1, pink=P2. The two large mink AY.39 clades are highlighted in blue (Clade A) and orange (Clade B) boxes. Spike reversion K478T is denoted by a star on the branch where it emerged while the samples harboring the reversion are indicated by blue triangle.

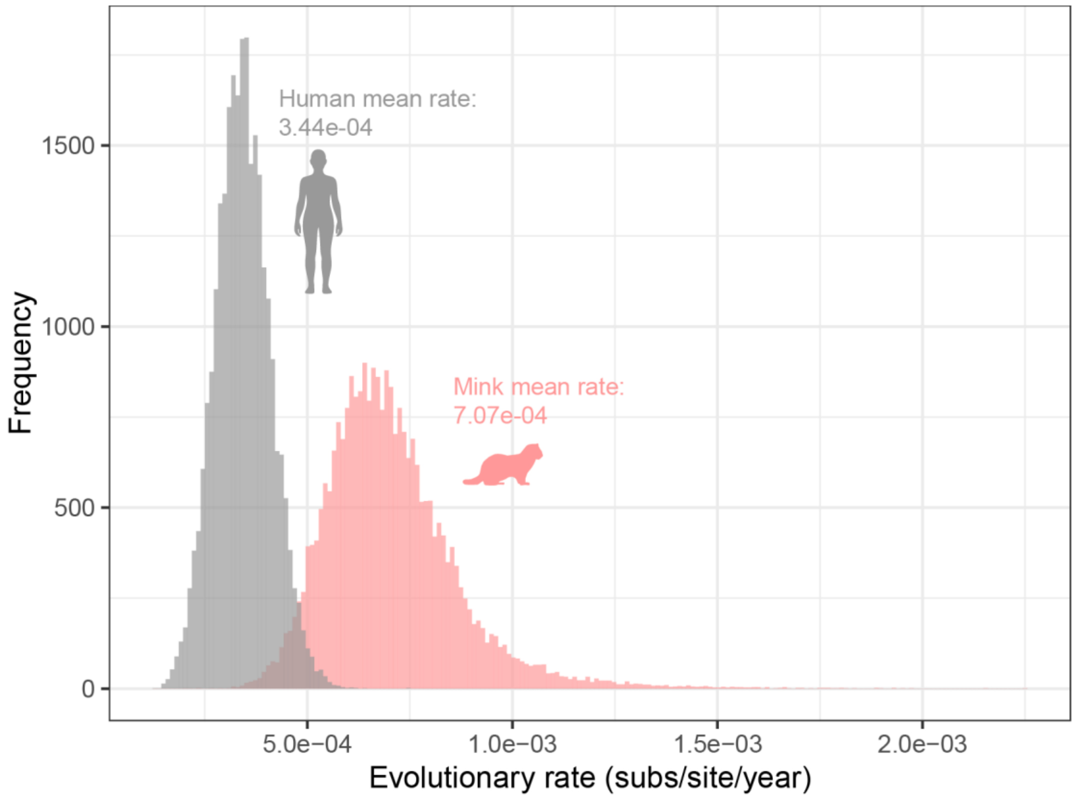

Figure S8: Evolutionary rates of Omicron BF.1 in humans and mink. Posterior distributions of evolutionary rates (substitutions per site per year) for whole-genome SARS-CoV-2 Omicron BF.1 variant sequences isolated from humans (grey) and mink (pink). Source data are provided in a Source Data file.

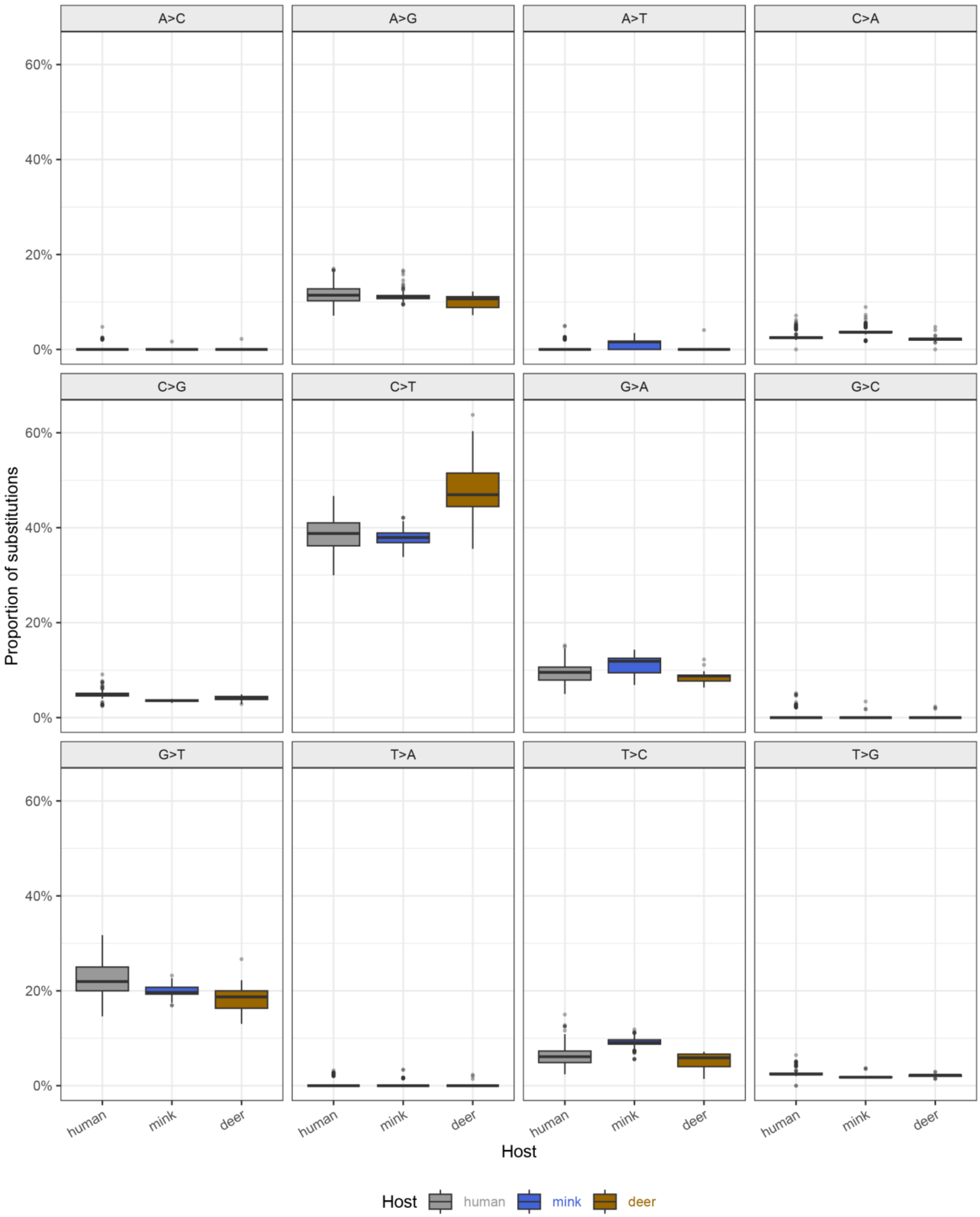

Figure S9. AY.39 mutational spectra by substitution type and host group. Boxplots of all 12 substitution type proportions in lineage AY.39 across different host group: Farm P mink (*n* = 176), WTD (*n* = 24) and humans (*n* = 389). Source data are provided in a Source Data file.

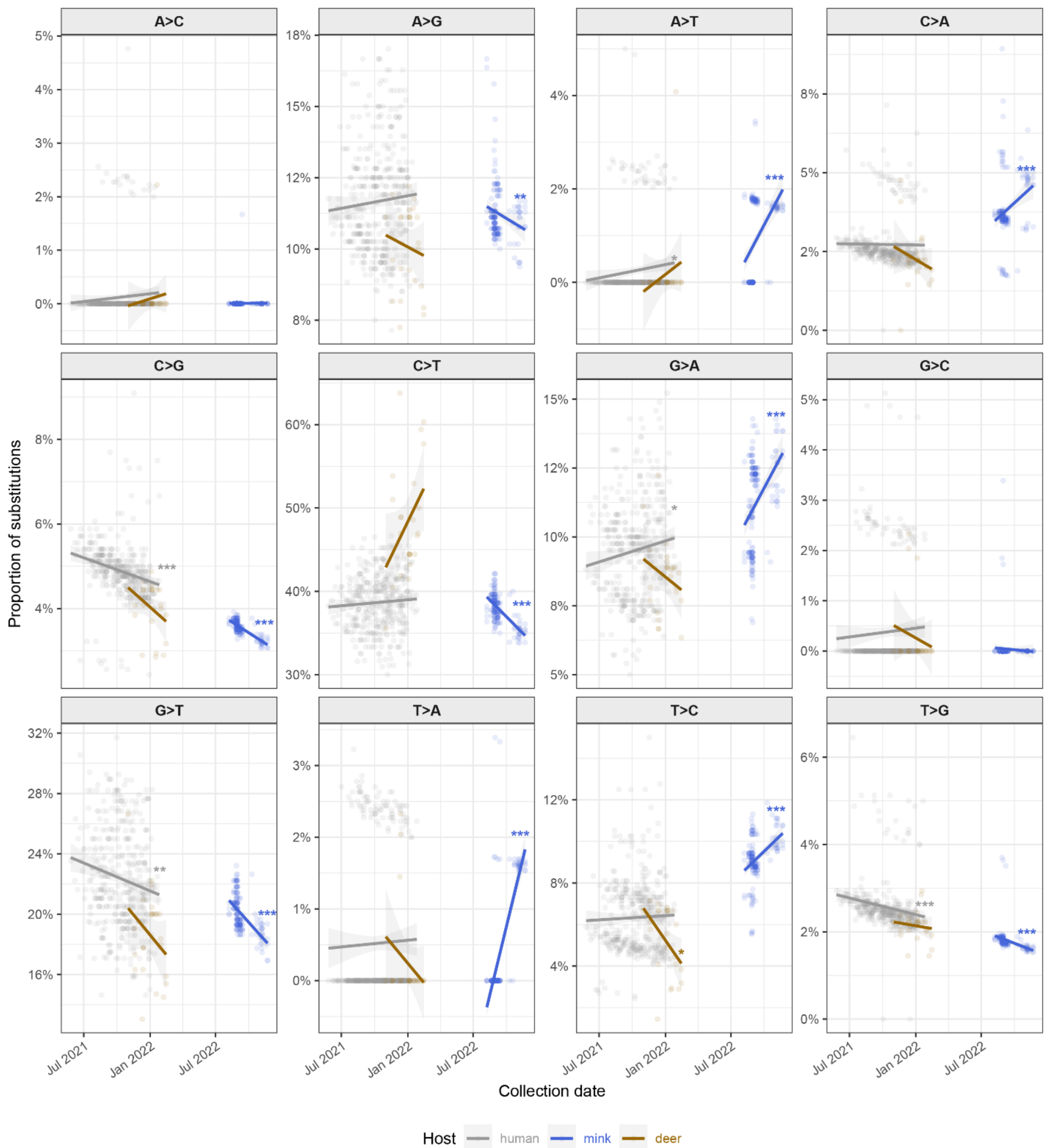

Figure S10. Temporal slopes of all 12 substitution type proportions by host group. Per-host linear regression lines with 95% confidence intervals are shown for each substitution type across AY.39 sequences from humans (grey), mink (blue), and WTD (brown). Slopes were estimated using ordinary least-squares linear regression. Asterisks at the end of each regression line indicate the significance of two-sided tests of the regression slope coefficient (* $p<0.05$, ** $p<0.01$, *** $p<0.001$). Source data are provided in a Source Data file. Full slope estimates, *p*-values, and R^2 values for all substitution types and host groups are reported in **Supplementary File 2**.

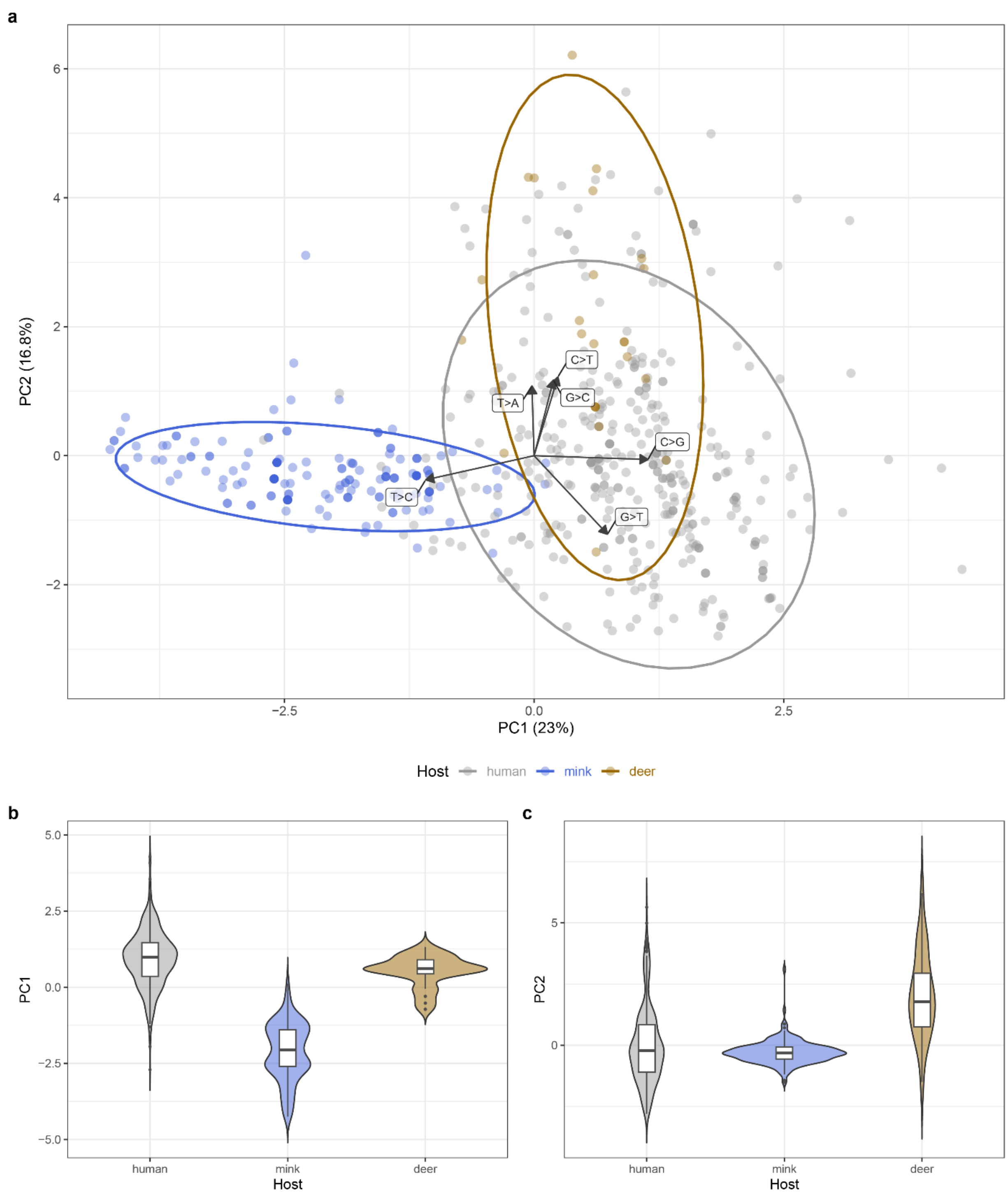

Figure S11: Principal components analysis of AY.39 mutation spectra. (a) Biplot of PC1 (23%) and PC2 (16.8%) with 95% confidence ellipses per host group. Arrows indicate the contribution of the six substitution types with the strongest loadings to the first two principal components. Distribution of (b) PC1 and (c) PC2 scores by host group shown as violin plots with embedded boxplots. Source data are provided in a Source Data file.

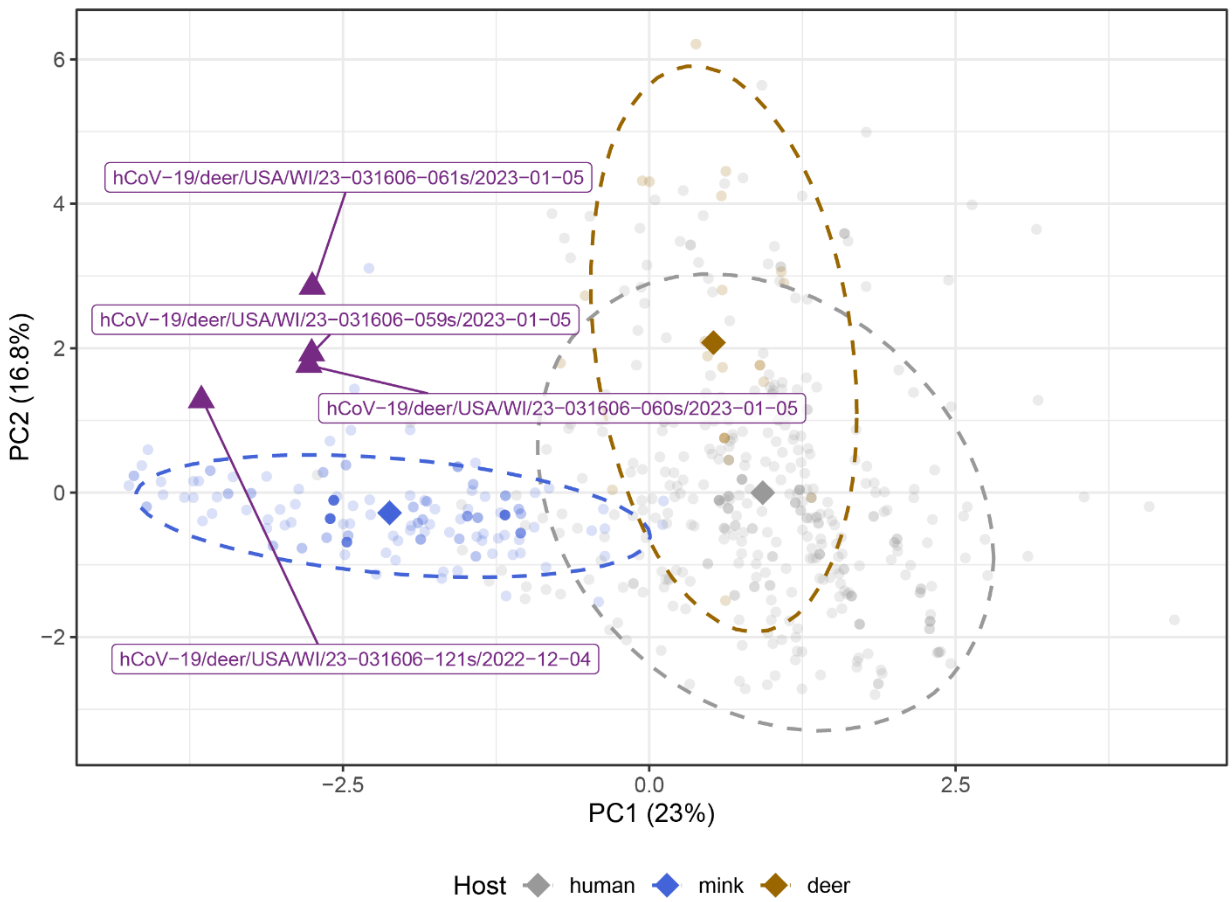

Figure S12: Mink-derived deer sequences projected into PCA space. The four white-tailed deer sequences phylogenetically nested within the mink clade (purple triangles) were excluded from the main PCA and projected post-hoc using the loadings derived from the main analysis. Dashed ellipses show 95% confidence intervals for each host group; diamonds indicate group centroids. Source data are provided in a Source Data file.

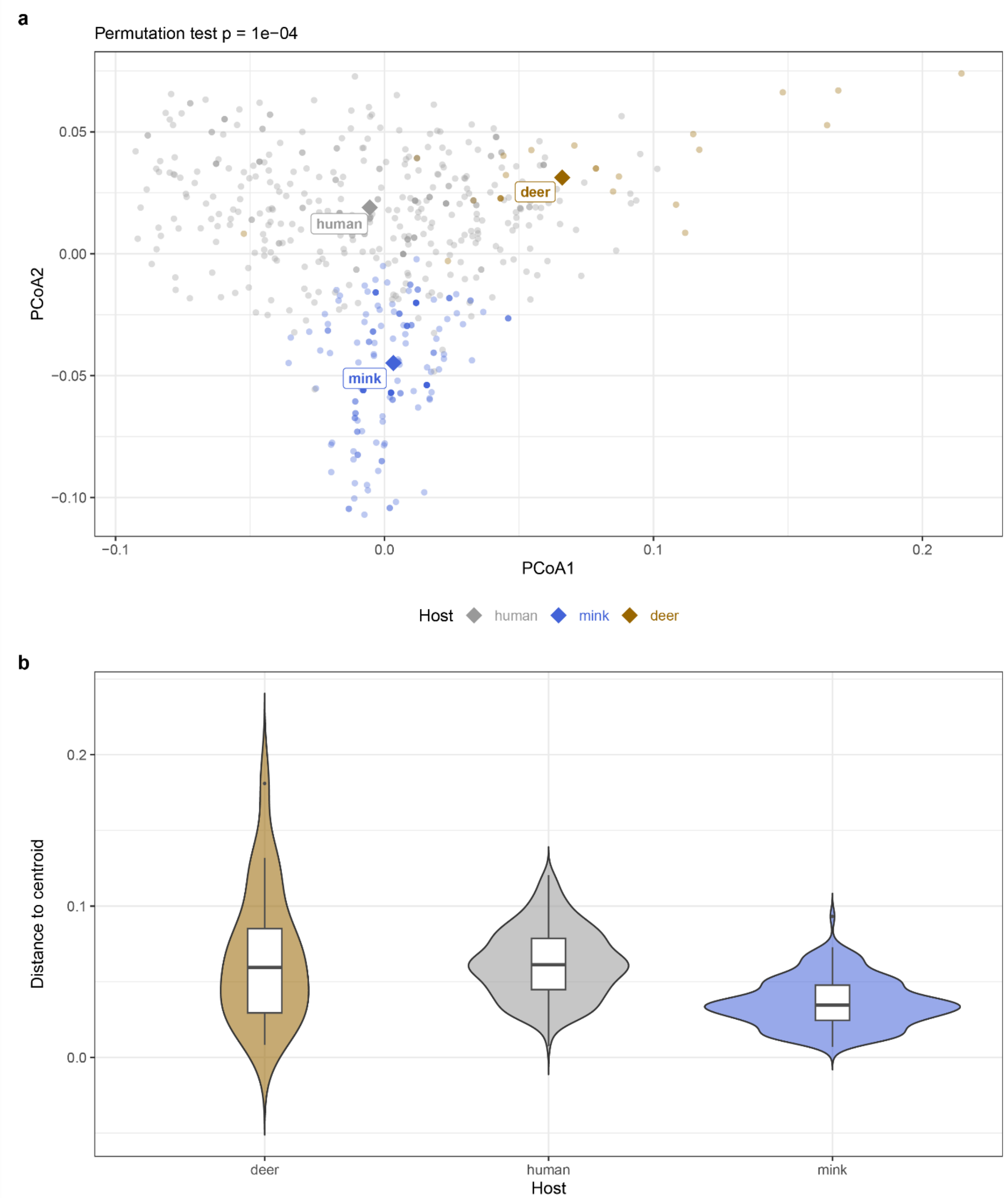

Figure S13. Within-group multivariate dispersion of AY.39 mutation spectra. (A) Principal coordinates analysis (PCoA) of Bray-Curtis dissimilarities between sequences, with group centroids indicated by diamond symbols. (B) Distances of individual sequences to their respective host-group centroid in Bray-Curtis dissimilarity space. Violin plots show the distribution of distances; center lines indicate the median and box bounds represent the interquartile range. Differences in dispersion were evaluated using a two-sided permutation test for homogeneity of multivariate dispersions (permutations n = 9,999), yielding a significant host effect ($p = 1 \times 10^{-4}$). No adjustment for multiple comparisons was applied. Source data are provided in a Source Data file.

Figure S14: Number of PCR test results and clade classification for samples obtained from individual pens from farm P (a) barn P1 and (b) barn P2. Each pen held the same individual(s) throughout the sample collection period. Sequenced isolates and their clade designation are depicted by colored circles (clade A=blue and clade B=orange). Source data are provided in a Source Data file.

Table S1: Number of SARS-CoV-2 outbreaks in mink reported to the Program for Monitoring Emerging Diseases (ProMED) and/or the World Animal Health Information System (WAHIS), number of mink-associated sequences available in GISAID by country (n=1,224) and US state (n=281), and number of sequences collected in this study by farm (n=293)

Country	Reported outbreaks[a]	Sequence collection year range	Pre-WHO classification sequences	Alpha sequences	Delta sequences	Omicron sequences	Total # of sequences
Belarus	-	2020	1	-	-	-	1
Bulgaria	1	2023	-	-	-	6	6
Latvia	1	2020–2022	187	-	12	-	199
France	2	2020	12	-	-	-	12
Canada	3	2020	5	-	-	-	5
Italy	4	2023	-	-	-	1	1
Denmark	8	2020	468	-	-	-	468
Lithuania	17	2020–2021	23	5	33	-	61
Spain	18	2020–2021	55	18	1	-	74
Poland	19	2020–2023	24	1	17	6	48
Greece	25	-	-	-	-	-	-
Sweden	25	-	-	-	-	-	-
Netherlands	29	2020	349	-	-	-	349
USA							
Location masked	1[b]	2021	-	-	-	-	-
Michigan	1	2020	71	-	-	-	71
Oregon	1	2020	4	-	-	-	4
Wisconsin	3[c]	2020	52	-	-	-	52
Utah	12[d]	2020–2021	154	-	-	-	154
This study							
Farm N (State 1)	-	2022	-	-	-	48	48
Farm O (State 1)	-	2022	-	-	-	1	1
Farm P (State 2)	-	2022	-	-	244	-	244

[a] Source: SARS-ANI project (https://github.com/amel-github/sars-ani)

[b] Outbreak detected through antibody testing; no sequences available

[c] The sole Wisconsin 2022 outbreak was detected through antibody testing; no sequences available

Table S2: Summary of diagnostic PCR test results for samples collected in each of the three farms with identified outbreaks in this study

Farm	# of mink in the farm	# of tested mink[a]	# of PCR positive mink[b]	# of total collected samples	# of negative samples	# of 'one gene' positive samples[c]	# of positive samples	Total positive samples (%)[d]
Farm N (State 1)	4,000	250	159	4,945	4,462	245	238	4.81%
Farm O (State 1)	14,000	250	5	3,685	3,655	25	5	0.14%
Farm P (State 2)	3,000	250	228	4,296	3,684	95	517	12.0%
all	21,000	750	392	12,926	11,801	365	760	5.88%

[a] At the start of the sampling period; some subjects were not tested at different time points because they were either culled or some other extenuating circumstance did not allow sample collection

[b] At any time in the sample collection period

[c] Positive for the N1/N2 regions of the SARS-CoV-2 N gene only

[d] Calculated using the total number of positive samples (excluding one gene positive samples)

Table S3: Whole genome sequence data for samples collected in this study

Sequence Name	State	Farm	Collection date	Pango lineage	VOC	GenBank Accession
hCoV-19/mink/USA/MN04N-212/2022	State 1	Farm N	2022-08-15	BA.5.2.1	Omicron	PQ594496
hCoV-19/mink/USA/MN08N-097/2022	State 1	Farm N	2022-09-13	BA.5.2.1	Omicron	PQ594511
hCoV-19/mink/USA/MN08N-104/2022	State 1	Farm N	2022-09-13	BA.5.2.1	Omicron	PQ594513
hCoV-19/mink/USA/MN08N-161/2022	State 1	Farm N	2022-09-13	BA.5.2.1	Omicron	PQ594515
hCoV-19/mink/USA/MN07N-189/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594497
hCoV-19/mink/USA/MN07N-195/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594498
hCoV-19/mink/USA/MN07N-224/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594499
hCoV-19/mink/USA/MN07N-225/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594500
hCoV-19/mink/USA/MN07N-226/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594501
hCoV-19/mink/USA/MN07N-227/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594502
hCoV-19/mink/USA/MN07N-229/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594503
hCoV-19/mink/USA/MN07N-231/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594504
hCoV-19/mink/USA/MN07N-237/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594505
hCoV-19/mink/USA/MN07N-239/2022	State 1	Farm N	2022-09-06	BF.1	Omicron	PQ594506
hCoV-19/mink/USA/MN08N-027/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594507
hCoV-19/mink/USA/MN08N-063/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594508
hCoV-19/mink/USA/MN08N-091/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594509
hCoV-19/mink/USA/MN08N-096/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594510
hCoV-19/mink/USA/MN08N-100/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594512
hCoV-19/mink/USA/MN08N-152/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594514
hCoV-19/mink/USA/MN08N-166/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594516
hCoV-19/mink/USA/MN08N-177/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594517
hCoV-19/mink/USA/MN08N-180/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594518
hCoV-19/mink/USA/MN08N-190/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594519
hCoV-19/mink/USA/MN08N-191/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594520
hCoV-19/mink/USA/MN08N-194/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594521
hCoV-19/mink/USA/MN08N-196/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594522
hCoV-19/mink/USA/MN08N-198/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594523
hCoV-19/mink/USA/MN08N-200/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594524
hCoV-19/mink/USA/MN08N-203/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594525
hCoV-19/mink/USA/MN08N-213/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594526
hCoV-19/mink/USA/MN08N-215/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594527
hCoV-19/mink/USA/MN08N-223/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594528
hCoV-19/mink/USA/MN08N-227/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594529
hCoV-19/mink/USA/MN08N-228/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594530
hCoV-19/mink/USA/MN08N-229/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594531
hCoV-19/mink/USA/MN08N-235/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594532
hCoV-19/mink/USA/MN08N-236/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594533
hCoV-19/mink/USA/MN08N-238/2022	State 1	Farm N	2022-09-13	BF.1	Omicron	PQ594534
hCoV-19/mink/USA/MN09N-005/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594535
hCoV-19/mink/USA/MN09N-010/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594536
hCoV-19/mink/USA/MN09N-157/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594537

hCoV-19/mink/USA/MN09N-177/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594538
hCoV-19/mink/USA/MN09N-181/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594539
hCoV-19/mink/USA/MN09N-204/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594540
hCoV-19/mink/USA/MN09N-211/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594541
hCoV-19/mink/USA/MN09N-221/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594542
hCoV-19/mink/USA/MN09N-240/2022	State 1	Farm N	2022-09-21	BF.1	Omicron	PQ594543
hCoV-19/mink/USA/MO08N-203/2022	State 1	Farm O	2022-09-12	BA.4.1	Omicron	PQ594544
hCoV-19/mink/USA/MP01N-123/2022	State 2	Farm P	2022-08-08	AY.39	Delta	PQ594545
hCoV-19/mink/USA/MP01N-150/2022	State 2	Farm P	2022-08-08	AY.39	Delta	PQ594546
hCoV-19/mink/USA/MP01N-151/2022	State 2	Farm P	2022-08-08	AY.39	Delta	PQ594547
hCoV-19/mink/USA/MP01N-239/2022	State 2	Farm P	2022-08-08	AY.39	Delta	PQ594548
hCoV-19/mink/USA/MP02N-025/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594549
hCoV-19/mink/USA/MP02N-059/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594550
hCoV-19/mink/USA/MP02N-066/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594551
hCoV-19/mink/USA/MP02N-067/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594552
hCoV-19/mink/USA/MP02N-158/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594553
hCoV-19/mink/USA/MP02N-163/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594554
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hCoV-19/mink/USA/MP02N-205/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594560
hCoV-19/mink/USA/MP02N-206/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594561
hCoV-19/mink/USA/MP02N-218/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594562
hCoV-19/mink/USA/MP02N-232/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594563
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hCoV-19/mink/USA/MP02N-237/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594566
hCoV-19/mink/USA/MP02N-245/2022	State 2	Farm P	2022-08-21	AY.39	Delta	PQ594567
hCoV-19/mink/USA/MP03N-002/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594568
hCoV-19/mink/USA/MP03N-005/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594569
hCoV-19/mink/USA/MP03N-007/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594570
hCoV-19/mink/USA/MP03N-017/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594571
hCoV-19/mink/USA/MP03N-018/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594572
hCoV-19/mink/USA/MP03N-022/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594573
hCoV-19/mink/USA/MP03N-023/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594574
hCoV-19/mink/USA/MP03N-026/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594575
hCoV-19/mink/USA/MP03N-027/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594576
hCoV-19/mink/USA/MP03N-039/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594577
hCoV-19/mink/USA/MP03N-042/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594578
hCoV-19/mink/USA/MP03N-043/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594579
hCoV-19/mink/USA/MP03N-051/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594580
hCoV-19/mink/USA/MP03N-056/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594581
hCoV-19/mink/USA/MP03N-060/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594582

hCoV-19/mink/USA/MP03N-069/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594583
hCoV-19/mink/USA/MP03N-071/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594584
hCoV-19/mink/USA/MP03N-075/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594585
hCoV-19/mink/USA/MP03N-076/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594586
hCoV-19/mink/USA/MP03N-103/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594587
hCoV-19/mink/USA/MP03N-118/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594588
hCoV-19/mink/USA/MP03N-149/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594589
hCoV-19/mink/USA/MP03N-152/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594590
hCoV-19/mink/USA/MP03N-157/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594591
hCoV-19/mink/USA/MP03N-159/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594592
hCoV-19/mink/USA/MP03N-160/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594593
hCoV-19/mink/USA/MP03N-170/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594594
hCoV-19/mink/USA/MP03N-172/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594595
hCoV-19/mink/USA/MP03N-173/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594596
hCoV-19/mink/USA/MP03N-175/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594597
hCoV-19/mink/USA/MP03N-176/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594598
hCoV-19/mink/USA/MP03N-182/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594599
hCoV-19/mink/USA/MP03N-185/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594600
hCoV-19/mink/USA/MP03N-189/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594601
hCoV-19/mink/USA/MP03N-193/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594602
hCoV-19/mink/USA/MP03N-194/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594603
hCoV-19/mink/USA/MP03N-195/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594604
hCoV-19/mink/USA/MP03N-197/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594605
hCoV-19/mink/USA/MP03N-198/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594606
hCoV-19/mink/USA/MP03N-199/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594607
hCoV-19/mink/USA/MP03N-200/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594608
hCoV-19/mink/USA/MP03N-201/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594609
hCoV-19/mink/USA/MP03N-202/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594610
hCoV-19/mink/USA/MP03N-203/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594611
hCoV-19/mink/USA/MP03N-204/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594612
hCoV-19/mink/USA/MP03N-207/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594613
hCoV-19/mink/USA/MP03N-209/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594614
hCoV-19/mink/USA/MP03N-210/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594615
hCoV-19/mink/USA/MP03N-211/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594616
hCoV-19/mink/USA/MP03N-216/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594617
hCoV-19/mink/USA/MP03N-217/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594618
hCoV-19/mink/USA/MP03N-219/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594619
hCoV-19/mink/USA/MP03N-220/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594620
hCoV-19/mink/USA/MP03N-223/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594621
hCoV-19/mink/USA/MP03N-224/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594622
hCoV-19/mink/USA/MP03N-225/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594623
hCoV-19/mink/USA/MP03N-228/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594624
hCoV-19/mink/USA/MP03N-231/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594625
hCoV-19/mink/USA/MP03N-235/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594626
hCoV-19/mink/USA/MP03N-241/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594627

hCoV-19/mink/USA/MP03N-243/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594628
hCoV-19/mink/USA/MP03N-244/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594629
hCoV-19/mink/USA/MP03N-248/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594630
hCoV-19/mink/USA/MP03N-249/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594631
hCoV-19/mink/USA/MP03N-250/2022	State 2	Farm P	2022-08-28	AY.39	Delta	PQ594632
hCoV-19/mink/USA/MP04N-002/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594633
hCoV-19/mink/USA/MP04N-003/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594634
hCoV-19/mink/USA/MP04N-005/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594635
hCoV-19/mink/USA/MP04N-007/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594636
hCoV-19/mink/USA/MP04N-008/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594637
hCoV-19/mink/USA/MP04N-011/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594638
hCoV-19/mink/USA/MP04N-017/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594639
hCoV-19/mink/USA/MP04N-018/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594640
hCoV-19/mink/USA/MP04N-019/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594641
hCoV-19/mink/USA/MP04N-021/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594642
hCoV-19/mink/USA/MP04N-022/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594643
hCoV-19/mink/USA/MP04N-029/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594644
hCoV-19/mink/USA/MP04N-032/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594645
hCoV-19/mink/USA/MP04N-033/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594646
hCoV-19/mink/USA/MP04N-036/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594647
hCoV-19/mink/USA/MP04N-038/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594648
hCoV-19/mink/USA/MP04N-039/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594649
hCoV-19/mink/USA/MP04N-042/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594650
hCoV-19/mink/USA/MP04N-043/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594651
hCoV-19/mink/USA/MP04N-046/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594652
hCoV-19/mink/USA/MP04N-050/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594653
hCoV-19/mink/USA/MP04N-051/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594654
hCoV-19/mink/USA/MP04N-052/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594655
hCoV-19/mink/USA/MP04N-054/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594656
hCoV-19/mink/USA/MP04N-056/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594657
hCoV-19/mink/USA/MP04N-063/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594658
hCoV-19/mink/USA/MP04N-071/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594659
hCoV-19/mink/USA/MP04N-072/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594660
hCoV-19/mink/USA/MP04N-073/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594661
hCoV-19/mink/USA/MP04N-075/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594662
hCoV-19/mink/USA/MP04N-076/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594663
hCoV-19/mink/USA/MP04N-077/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594664
hCoV-19/mink/USA/MP04N-083/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594665
hCoV-19/mink/USA/MP04N-084/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594666
hCoV-19/mink/USA/MP04N-089/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594667
hCoV-19/mink/USA/MP04N-092/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594668
hCoV-19/mink/USA/MP04N-098/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594669
hCoV-19/mink/USA/MP04N-101/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594670
hCoV-19/mink/USA/MP04N-102/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594671
hCoV-19/mink/USA/MP04N-103/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594672

hCoV-19/mink/USA/MP04N-104/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594673
hCoV-19/mink/USA/MP04N-108/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594674
hCoV-19/mink/USA/MP04N-112/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594675
hCoV-19/mink/USA/MP04N-115/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594676
hCoV-19/mink/USA/MP04N-117/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594677
hCoV-19/mink/USA/MP04N-148/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594678
hCoV-19/mink/USA/MP04N-149/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594679
hCoV-19/mink/USA/MP04N-160/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594680
hCoV-19/mink/USA/MP04N-170/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594681
hCoV-19/mink/USA/MP04N-174/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594682
hCoV-19/mink/USA/MP04N-175/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594683
hCoV-19/mink/USA/MP04N-176/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594684
hCoV-19/mink/USA/MP04N-182/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594685
hCoV-19/mink/USA/MP04N-186/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594686
hCoV-19/mink/USA/MP04N-193/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594687
hCoV-19/mink/USA/MP04N-194/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594688
hCoV-19/mink/USA/MP04N-200/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594689
hCoV-19/mink/USA/MP04N-201/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594690
hCoV-19/mink/USA/MP04N-203/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594691
hCoV-19/mink/USA/MP04N-209/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594692
hCoV-19/mink/USA/MP04N-210/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594693
hCoV-19/mink/USA/MP04N-211/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594694
hCoV-19/mink/USA/MP04N-216/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594695
hCoV-19/mink/USA/MP04N-217/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594696
hCoV-19/mink/USA/MP04N-230/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594697
hCoV-19/mink/USA/MP04N-241/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594698
hCoV-19/mink/USA/MP04N-243/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594699
hCoV-19/mink/USA/MP04N-249/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594700
hCoV-19/mink/USA/MP04N-250/2022	State 2	Farm P	2022-08-31	AY.39	Delta	PQ594701
hCoV-19/mink/USA/MP05N-002/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594702
hCoV-19/mink/USA/MP05N-006/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594703
hCoV-19/mink/USA/MP05N-015/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594704
hCoV-19/mink/USA/MP05N-028/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594705
hCoV-19/mink/USA/MP05N-029/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594706
hCoV-19/mink/USA/MP05N-030/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594707
hCoV-19/mink/USA/MP05N-035/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594708
hCoV-19/mink/USA/MP05N-040/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594709
hCoV-19/mink/USA/MP05N-044/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594710
hCoV-19/mink/USA/MP05N-045/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594711
hCoV-19/mink/USA/MP05N-047/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594712
hCoV-19/mink/USA/MP05N-048/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594713
hCoV-19/mink/USA/MP05N-049/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594714
hCoV-19/mink/USA/MP05N-050/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594715
hCoV-19/mink/USA/MP05N-053/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594716
hCoV-19/mink/USA/MP05N-065/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594717

hCoV-19/mink/USA/MP05N-068/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594718
hCoV-19/mink/USA/MP05N-072/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594719
hCoV-19/mink/USA/MP05N-079/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594720
hCoV-19/mink/USA/MP05N-080/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594721
hCoV-19/mink/USA/MP05N-081/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594722
hCoV-19/mink/USA/MP05N-082/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594723
hCoV-19/mink/USA/MP05N-085/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594724
hCoV-19/mink/USA/MP05N-086/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594725
hCoV-19/mink/USA/MP05N-087/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594726
hCoV-19/mink/USA/MP05N-088/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594727
hCoV-19/mink/USA/MP05N-093/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594728
hCoV-19/mink/USA/MP05N-094/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594729
hCoV-19/mink/USA/MP05N-095/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594730
hCoV-19/mink/USA/MP05N-096/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594731
hCoV-19/mink/USA/MP05N-097/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594732
hCoV-19/mink/USA/MP05N-098/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594733
hCoV-19/mink/USA/MP05N-099/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594734
hCoV-19/mink/USA/MP05N-101/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594735
hCoV-19/mink/USA/MP05N-105/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594736
hCoV-19/mink/USA/MP05N-106/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594737
hCoV-19/mink/USA/MP05N-107/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594738
hCoV-19/mink/USA/MP05N-108/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594739
hCoV-19/mink/USA/MP05N-109/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594740
hCoV-19/mink/USA/MP05N-111/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594741
hCoV-19/mink/USA/MP05N-113/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594742
hCoV-19/mink/USA/MP05N-114/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594743
hCoV-19/mink/USA/MP05N-116/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594744
hCoV-19/mink/USA/MP05N-119/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594745
hCoV-19/mink/USA/MP05N-120/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594746
hCoV-19/mink/USA/MP05N-125/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594747
hCoV-19/mink/USA/MP05N-130/2022	State 2	Farm P	2022-09-07	AY.39	Delta	PQ594748
hCoV-19/mink/USA/MP06N-040/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594749
hCoV-19/mink/USA/MP06N-041/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594750
hCoV-19/mink/USA/MP06N-082/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594751
hCoV-19/mink/USA/MP06N-086/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594752
hCoV-19/mink/USA/MP06N-091/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594753
hCoV-19/mink/USA/MP06N-105/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594754
hCoV-19/mink/USA/MP06N-113/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594755
hCoV-19/mink/USA/MP06N-122/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594756
hCoV-19/mink/USA/MP06N-129/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594757
hCoV-19/mink/USA/MP06N-130/2022	State 2	Farm P	2022-09-13	AY.39	Delta	PQ594758
hCoV-19/mink/USA/MP11N-050/2022	State 2	Farm P	2022-10-10	AY.39	Delta	PQ594759
hCoV-19/mink/USA/MP11N-210/2022	State 2	Farm P	2022-10-10	AY.39	Delta	PQ594760
hCoV-19/mink/USA/MP12N-175/2022	State 2	Farm P	2022-10-19	AY.39	Delta	PQ594761
hCoV-19/mink/USA/MP13N-192/2022	State 2	Farm P	2022-10-24	AY.39	Delta	PQ594762

hCoV-19/mink/USA/MP13N-193/2022	State 2	Farm P	2022-10-24	AY.39	Delta	PQ594763
hCoV-19/mink/USA/MP13N-214/2022	State 2	Farm P	2022-10-24	AY.39	Delta	PQ594764
hCoV-19/mink/USA/MP13N-225/2022	State 2	Farm P	2022-10-24	AY.39	Delta	PQ594765
hCoV-19/mink/USA/MP14N-157/2022	State 2	Farm P	2022-11-03	AY.39	Delta	PQ594766
hCoV-19/mink/USA/MP14N-173/2022	State 2	Farm P	2022-11-03	AY.39	Delta	PQ594767
hCoV-19/mink/USA/MP14N-188/2022	State 2	Farm P	2022-11-03	AY.39	Delta	PQ594768
hCoV-19/mink/USA/MP14N-198/2022	State 2	Farm P	2022-11-03	AY.39	Delta	PQ594769
hCoV-19/mink/USA/MP14N-223/2022	State 2	Farm P	2022-11-03	AY.39	Delta	PQ594770
hCoV-19/mink/USA/MP14N-230/2022	State 2	Farm P	2022-11-03	AY.39	Delta	PQ594771
hCoV-19/mink/USA/MP15N-157/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594772
hCoV-19/mink/USA/MP15N-158/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594773
hCoV-19/mink/USA/MP15N-161/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594774
hCoV-19/mink/USA/MP15N-174/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594775
hCoV-19/mink/USA/MP15N-175/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594776
hCoV-19/mink/USA/MP15N-195/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594777
hCoV-19/mink/USA/MP15N-198/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594778
hCoV-19/mink/USA/MP15N-219/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594779
hCoV-19/mink/USA/MP15N-223/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594780
hCoV-19/mink/USA/MP15N-233/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594781
hCoV-19/mink/USA/MP15N-235/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594782
hCoV-19/mink/USA/MP15N-245/2022	State 2	Farm P	2022-11-07	AY.39	Delta	PQ594783
hCoV-19/mink/USA/MP16N-159/2022	State 2	Farm P	2022-11-22	AY.39	Delta	PQ594784
hCoV-19/mink/USA/MP16N-160/2022	State 2	Farm P	2022-11-22	AY.39	Delta	PQ594785
hCoV-19/mink/USA/MP16N-165/2022	State 2	Farm P	2022-11-22	AY.39	Delta	PQ594786
hCoV-19/mink/USA/MP16N-166/2022	State 2	Farm P	2022-11-22	AY.39	Delta	PQ594787
hCoV-19/mink/USA/MP16N-170/2022	State 2	Farm P	2022-11-22	AY.39	Delta	PQ594788

Table S4: Proportion of SARS-CoV-2 virus lineages corresponding to GISAID samples collected from mink (n=1,505)

Rank	Lineage	Percentage	Number	Variant
1	B.1.1.298	28.9%	434	Pre-WHO
2	B.1	21.7%	327	Pre-WHO
3	B.1.8	15.4%	232	Pre-WHO
4	B.1.177.60	12.9%	194	Pre-WHO
5	B.1.22	4.5%	68	Pre-WHO
6	B.11	2.9%	44	Pre-WHO
7	B.1.1	1.4%	21	Pre-WHO
9	B.1.1.7	1.3%	20	Alpha
8	AY.4.5	1.3%	20	Delta
10	B.1.177	0.9%	13	Pre-WHO
11	B.1.160	0.8%	12	Pre-WHO
12	B.1.536	0.7%	10	Pre-WHO
13	B.1.1.280	0.6%	9	Pre-WHO
14	AY.4	0.5%	8	Delta
15	AY.43	0.5%	8	Delta
16	B.1.1.307	0.5%	8	Pre-WHO
18	B.1.617.2	0.5%	7	Delta
17	B.1.1.219	0.5%	7	Pre-WHO
19	AY.122	0.4%	6	Delta
20	B.1.343	0.4%	6	Pre-WHO
21	AW.1	0.3%	5	Pre-WHO
24	Q.1	0.3%	4	Alpha
23	BA.5.2	0.3%	4	Omicron
22	B.1.2	0.3%	4	Pre-WHO
25	AY.126	0.2%	3	Delta
26	AY.4.2	0.2%	3	Delta
29	XBB.1.22	0.2%	3	Omicron
27	B.1.1.170	0.2%	3	Pre-WHO
28	B.1.1.294	0.2%	3	Pre-WHO
30	AY.33	0.1%	2	Delta
31	AY.6	0.1%	2	Delta
34	BA.2	0.1%	2	Omicron
35	FY.1.2	0.1%	2	Omicron
32	B	0.1%	2	Pre-WHO
33	B.1.258.9	0.1%	2	Pre-WHO
36	AY.121	0.1%	1	Delta
37	AY.4.3	0.1%	1	Delta
38	AY.5	0.1%	1	Delta
39	AY.98.1	0.1%	1	Delta
40	FY.1	0.1%	1	Omicron
41	XBB.1.5	0.1%	1	Omicron
42	Unassigned	0.1%	1	NA

Table S5: Prevalence of amino acid changes present in >10% of Farm P samples among complete SARS-CoV-2 genomic sequences available in GISAID

Amino acid change (nucleotide mutation)	Prevalence in GISAID human sequences (n=15,300,583)	Prevalence in Delta GISAID human sequences (n=4,300,059)	Prevalence in AY.39 GISAID human sequences (n=51,560)	Prevalence in GISAID mink (n=1,442)
NSP2:G3339S (G1820A)	129,726 (0.85%)	67,084 (1.56%)	732 (1.42%)	5 (0.29%)
NSP3:Y1185N (T6272A)	5,520 (<0.05%)	674 (<0.05%)	None (0%)	None (0%)
NSP4:G196S (G9140A)	824 (<0.05%)	145 (<0.05%)	1 (<0.05%)	None (0%)
NSP8:T187I (C12651T)	23,673 (0.15%)	6,339 (0.15%)	14 (<0.05%)	None (0%)
NSP9:G37E (G12795A)	390 (<0.05%)	292 (<0.05%)	None (0%)	305 (21.1%)
NSP9:P57H (C12855A)	405 (<0.05%)	85 (<0.05%)	8 (<0.05%)	7 (0.48%)
S:K478T (A22995C)	NA	NA	NA	NA
S:Y453F (A22920T)	2,835 (<0.05%)	202 (<0.05%)	None (0%)	624 (43.3%)
ORF3a:V55I (G25555A)	1,508 (<0.05%)	391 (<0.05%)	2 (<0.05%)	2 (0.14%)
ORF8:I10T (T27922C)	5,188 (<0.05%)	1,020 (<0.05%)	9 (<0.05%)	None (0%)
ORF10:T38I (C29670T)	11,929 (0.08%)	8,309 (0.19%)	13 (<0.05%)	None (0%)

Table S6: EPI_SET identifiers for GISAID sequences used in this study

EPI_SET Identifier	DOI link	Description	Acknowledgement Table	Analysis
EPI_SET_250414vh	10.55876/gis8.250414vh	Global mink sequences	Supplementary File 3	Phylogenetic analysis (Figure 3)
EPI_SET_241106uv	10.55876/gis8.241106uv	Global AY.39 sequences	Supplementary File 4	Phylogenetic analysis (Figures 5 and S5) Mutation spectrum analysis (Figures 6, S9-S13)
EPI_SET_241106cm	10.55876/gis8.241106cm	Global BF.1 sequences	Supplementary File 5	Phylogenetic analysis (Figure S4)